

MATH 4540 Final Project

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I. HYPERBOLIC PLANE GEOMETRY

A. Upper half-plane Model

We can begin looking at hyperbolic geometry by examining the pseudosphere and the upper-half plane model. In this model, we can parametrize the pseudosphere using

$$\tilde{\sigma}(v, w) = \left(\frac{1}{w} \cos v, \frac{1}{w} \sin v, \sqrt{1 - \frac{1}{w^2}} - \cosh^{-1} w \right).$$

This parametrization is only well defined and smooth for $w > 1$ due to the square-root in the parametrization. However, we can effectively ignore this boundary at $w = 1$ since the properties of surfaces only depend on the first fundamental form. We find the first fundamental form to be

$$FFF = \frac{dv^2 + dw^2}{w^2}$$

which is undefined when $w = 0$. Thus we confine ourselves to the upper half of the plane where $w > 0$ and geodesics are well defined, hence the name "upper half-plane model."

To see that this model is conformal, recall corollary 6.3.4 which says that a surface patch $\sigma(v, w)$ is conformal (preserves angles) if and only if the first fundamental form is of the form $\lambda(dv^2 + dw^2)$ where $\lambda(v, w)$ is smooth [1]. From our first fundamental form, we see that $\lambda(v, w) = \frac{1}{w^2}$ which is smooth for $w > 0$, so we have a conformal model.

Now we can consider the distances between two points $a, b \in \mathcal{H}$ which is given by

$$d_{\mathcal{H}}(a, b) = 2 \tanh^{-1} \left| \frac{b - a}{b - \bar{a}} \right|$$

where $\bar{a}$ is the complex conjugate of a . Here the $\tanh^{-1}$ function hints at the name "hyperbolic geometry."

We can also see the implications of the name "pseudosphere" by looking at the area of n -sided polygons on the surface of the pseudosphere. As shown in Figure 1, for the upper half-plane model, some hyperbolic lines (geodesics) look like semicircles, with their centers on the real axis. The other type of geodesics we call *half-lines* which extend upward from the real axis, parallel to the imaginary axis. For an n -sided polygon in $\mathcal{H}$ with

internal angles given by $\alpha_1, \alpha_2, \dots, \alpha_n$, the area is given by

$$\mathcal{A}(\mathcal{P}) = (n - 2)\pi - \sum_{i=1}^n \alpha_i.$$

This can be directly compared to the area of the same n -sided polygon on the sphere given by

$$\mathcal{A}(\mathcal{P}) = \sum_{i=1}^n \alpha_i - (n - 2)\pi.$$

As we can see, these two formulas differ by a negative sign, supporting the name "pseudosphere".

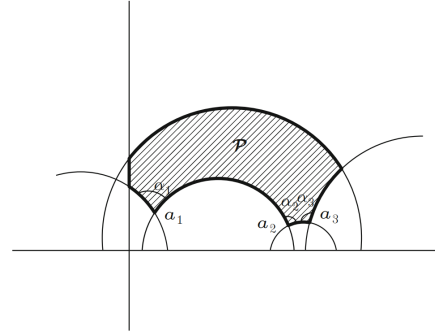


Figure 1. n -sided polygon in the upper half-plane model [1].

B. Poincaré disc Model

In addition to the upper half-plane model, we can also use the Poincaré disc model which uses the unit disc in the complex plane. We start with the transformation

$$\mathcal{P}(z) = \frac{z - i}{z + 1}, \quad z \in \mathbb{C}$$

which is well defined at all points of $\mathcal{H}$ and the boundary is the real axis. With this transformation, we can take $\mathcal{H}$ to the unit disc in the complex plane expressed as

$$\mathcal{D} = \{z \in \mathbb{C} \mid |z| < 1\}.$$

To ensure we are modeling hyperbolic geometry, we need $\mathcal{P} : \mathcal{H} \rightarrow \mathcal{D}_{\mathcal{P}}$ to be an isometry. The first fundamental form of $\mathcal{D}_{\mathcal{P}}$ is given by

$$FFF = \frac{4(dv^2 + dw^2)}{(1 - v^2 - w^2)^2}.$$

Then, if we let

$$\tilde{v} = \frac{-2w^2}{(v-1)^2 + w^2} \quad \text{and} \quad \tilde{w} = \frac{1 - v^2 - w^2}{(v-1)^2 + w^2},$$

we find that the first fundamental form can be reparametrized to be

$$FFF = \frac{d\tilde{v}^2 + d\tilde{w}^2}{\tilde{w}^2}.$$

This is the same first fundamental form as the upper half-plane model, meaning $\mathcal{D}_P$ is a conformal model of hyperbolic geometry.

Similar in form to the upper half-plane model, the distance between points $a, b \in \mathcal{D}_P$ is given by

$$d_{\mathcal{D}_P}(a, b) = 2 \tanh^{-1} \frac{|b - a|}{|1 - \bar{a}b|}.$$

In this model, we also see more reason for the name "pseudosphere". For a hyperbolic triangle of side lengths A, B, C and angles α, β, γ respectively, we have

$$\cosh C = \cosh A \cosh B - \sinh A \sinh B \cos \gamma.$$

This is very similar to the Euclidean law of cosines which for the same triangle states

$$C^2 = A^2 + B^2 - 2AB \cos \gamma,$$

and the law of cosines for spherical geometry,

$$\cos C = \cos A \cos B + \sin A \sin B \cos \gamma.$$

C. Isometries of $\mathcal{H}$

Just like the isometries of translations, rotations, and reflections on the Euclidean plane, there are a number of isometries for $\mathcal{H}$: First, the translation parallel to the real axis given by

$$T_a(z) = z + a, \quad a \in \mathbb{R}.$$

Second, the reflection in lines parallel to the imaginary axis given by

$$R_a(z) = 2a - \bar{z}, \quad a \in \mathbb{R}.$$

In other words, this reflects the point z over the half-line that intersects the real axis at a . Third, a dilation by a factor of $a > 0$ given by

$$D_a(z) = az.$$

The fourth isometry is less intuitive: The inversions in circles with centers on the real axis given by

$$\mathcal{I}_{a,r}(z) = a + \frac{r^2}{\bar{z} - a},$$

with the center of the circle being $a \in \mathbb{R}$ and radius $r > 0$. Additionally, any finite composition of these four isometries is also an isometry of $\mathcal{H}$.

We know that isometries take geodesics to geodesics. This means that the four isometries above take the semicircular geodesics and half-line geodesics to other semicircular or half-line geodesics. Intuitively, we can also see that translations, reflections, and dilations take semicircular geodesics to other semicircular geodesics and half-line geodesics to other half-line geodesics. This is not true for the fourth isometry involving the inversion of circles. Instead, if the inversion of the circle is centered at $a \in \mathbb{R}$ with $r > 0$, then all the hyperbolic geodesics that pass through the real axis at the point a are taken to half-line geodesics. All other geodesics are taken to semicircle geodesics.

From these isometries, we have a proposition that if l_1 and l_2 are hyperbolic lines in $\mathcal{H}$ with z_1 and z_2 being points on l_1 and l_2 respectively, then there is an isometry of $\mathcal{H}$ that takes l_1 to l_2 and z_1 to z_2 . To prove this, it is enough to show that this is true when $l_2 = l$ is the half line passing through the origin and $z_2 = z = i$. If we prove this, then we know there's an isometry from l_1 to l and z_1 to i as well as an isometry from l_2 to l and z_2 to i . Thus there is an isometry from l_1 to l_2 and from z_1 to z_2 . We start by considering two cases. First, start with l_1 as a half-line. We can use a translation (which is an isometry on $\mathcal{H}$) to take l_1 to l and z_1 to some point ib on l . Then we can use a dilation which is also an isometry to rescale ib to i . Thus the desired isometry is the composition of the translation and dilation. For case 2, l_1 is a semicircle. Here we can use the inversion isometry to take l_1 to a half-line geodesic, say l' . We now have case 1, so we know there is an isometry from l' to l . Thus the desired isometry is the composition of an inversion, a translation, and a dilation.

As an application to this proposition, we know that there is an isometry that takes a to b with $a, b \in \mathcal{H}$. Then that isometry will take the hyperbolic circle of radius $\mathcal{R}$ with center a to a circle of the same radius with center b . Additionally, we are able to see that in hyperbolic geometry, contrary to euclidean geometry, similar triangles are congruent. To prove this, we can define two triangles: One with vertices a, b, c and angles α, β, γ , and the other with vertices a', b', c' and angles α', β', γ' . Then if the corresponding angles of the triangles are equal ($\alpha = \alpha', \beta = \beta', \gamma = \gamma'$), we just need to show there is an isometry $\mathcal{F}$ of $\mathcal{H}$ such that $\mathcal{F}(a) = a', \mathcal{F}(b) = b', \mathcal{F}(c) = c'$.

D. Hyperbolic Geometry in Corals

Despite hyperbolic geometry being unintuitive to the human mind, it is still widely present in nature. One of the most common reasons for hyperbolic geometry to

arise is the desire for an organism to maximize surface area. This can be seen in corals who maximize their surface area in order to allow the greatest amount of algae to photosynthesize and produce energy for the coral.

Achieving this by utilizing hyperbolic geometry makes sense. In our every day lives, we see things like water droplets that use spherical geometry to *minimize* their surface area. In the exact opposite way, the geometry of the pseudosphere can be used to maximize surface area. If the surface wrinkles up on itself, there will be numerous folds and crevices to increase the surface area. This maximization of surface area of corals by using hyperbolic geometry can be seen in the crocheting of coral-like surfaces. The formula to crochet such a surface is to "crochet n stitches, increase one, repeat ad infinitum" [2]. A formula such as this steadily increases the number of stitches and thus the surface area of the coral. In creating this area-maximizing surface, we have a hyperbolic surface that curves away from itself, creating the characteristic folds and wrinkles that make corals so identifiable.

E. Hyperbolic Geometry in Sound Propagation

However, hyperbolic geometry in nature is not limited to the physical structure of organisms. Quite surprisingly, we see hyperbolic geometry appear in sound propagation. As humans, nearly all our experiences with sound occur on the surface of earth, with sound waves propagating through the air in the atmosphere. In these situations, we know sound to travel in straight lines from one point to another.

However, when we move from our atmosphere to the depths of the ocean, we do not see the same thing. Instead, we see that the speed of sound is a function of depth as shown in Figure 2. Near the surface, factors like water temperature and salinity are the dominating factors that determine the speed [3]. However, after a certain depth, it is the density of the water that is the dominating affect, and the speed of sound increases linearly with depth.

As a result, the fastest path for sound to take between two points in the ocean at large depths is not what humans would think of as the path of shortest distance. Instead, the fastest path is that along the circle whose center is on the surface of the ocean that contains both points. This has interesting implications, particularly for marine animals like whales. If whales are communicating with each other at these depths as shown in Figure 3, sounds travels along what we would consider "curved" lines. Then if a whale heard a distant sound coming from straight ahead of them, they would know that their fellow whale is actually *above* and in front of them, as shown by whales A and B in Figure 3.

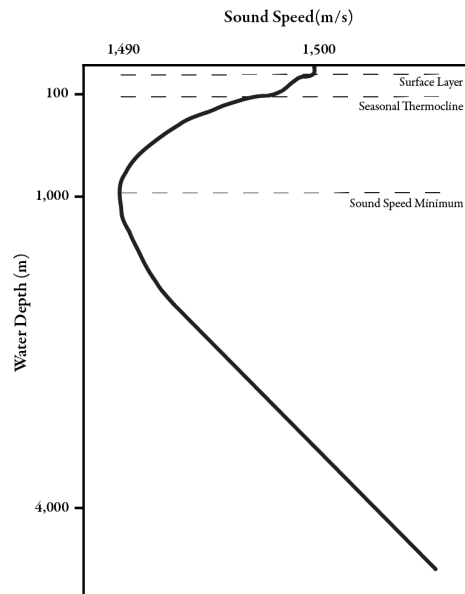


Figure 2. Speed of sound in the ocean as a function of depth [3].

We also know that this is hyperbolic geometry via the shape of the sound geodesics. In examining the upper half-plane model, we saw that geodesics take the form of half-lines parallel to the imaginary axis and semicircles with their centers on the real axis [1]. In this case, we clearly see that the sound geodesics form semicircles with their centers on the surface of the ocean. However, the semicircle geodesics lie below the surface of the ocean, rather than above as we would expect for the upper half-plane model. This means in the case of sound propagation like we see from Figure 3, we are using the *lower* half-plane model which is indeed hyperbolic geometry.

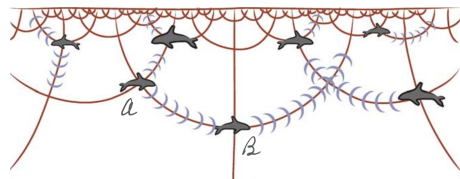


Figure 3. Path of sound for quickest communication between whales [4].

II. MINKOWSKI METRIC

To explore the applications of differential geometry, we can look at Einstein's theory of general relativity and how flat spacetime is described mathematically. When spacetime is flat, that means there is no mass

present. In this case, spacetime is described by a pseudo-riemannian metric defined as a "a smooth symmetric 2-tensor field g that is nondegenerate at each point" with the same signature everywhere [5]. These pseudo-riemannian metrics take the form

$$\bar{q}^{(r,s)} = (dx^1)^2 + \dots + (dx^r)^2 - (d\tau^1)^2 - \dots - (d\tau^s)^2.$$

From this definition, we can look at *Lorentz metrics* which are defined by to have an index of 1. To be more specific, we can look at the particular Lorentz metric

$$\bar{q}^{(r,1)} = (dx^1)^2 + \dots + (dx^r)^2 - (d\tau^1)^2$$

which we call the Minkowski metric. As we can see, the Minkowski metric describes a $(r+1)$ dimensional space.

It is only natural for Einstein's theory of general relativity to consider the Minkowski metric when $r = 3$, giving us three spatial dimensions and one time dimension. With these dimensions, we find the Minkowski metric to be

$$\bar{q}^{(3,1)} = (dx^1)^2 + (dx^2)^2 + (dx^3)^2 - (d\tau^1)^2.$$

This metric is an invariant in general relativity, meaning that the laws of physics are the same in all systems where the metric takes the same form as above. However, in general relativity, it is most often expressed as either

$$ds^2 = -(cdt)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2$$

where ds is an infinitesimal length, or in matrix form

$$\eta_{\alpha\beta} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where the time component comes first.

A. Hyperboloid Model and the Minkowski metric

The Minkowski metric has interesting connections to hyperbolic geometry as well. To make these connections, we use a third model for hyperbolic geometry called the hyperboloid model. In short, the Minkowski metric is the standard metric for the upper sheet of a two-sheeted hyperboloid. To see how this arises, we can start with the standard two-sheeted hyperboloid in three dimensions given by $x^2 + y^2 - z^2 = -1$. We can then bump this up into an extra dimension to include time, yielding the quadratic form $q(x, y, z, t) = x^2 + y^2 + z^2 - t^2 = -1$ [6][7]. Now, we can take this into the metric form to give

$$\bar{q}^{(3,1)} = (dx)^2 + (dy)^2 + (dz)^2 - (dt)^2.$$

Restricting ourselves to $t > 0$, we find that the upper sheet hyperboloid model gives exactly the Minkowski metric [5]. And if $r = 3$ in $\bar{q}^{(r,1)}$, we get the Minkowski metric with units $c = 1$ relevant to general relativity.

B. Introducing mass

In general relativity, the Minkowski metric of the form

$$ds^2 = -(cdt)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2$$

is only for flat spacetime, meaning there is no mass present. To accurately take into account gravitational effects that warp spacetime, the metric, namely the Lorentz metric, needs to be able to change from point to point. The most simple case where spacetime is not flat is given by the Schwarzschild metric which describes space with a massive spherically symmetric object. Using spherical coordinates and units of $c = 1$ and the gravitational constant $G = 1$, the metric is given by

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

where M is the mass of the object. From this metric, we can clearly see the spherical symmetry in the $d\theta^2$ and $d\phi^2$ terms which have no mass dependence. The metric can also be expressed in matrix form, given by

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2M}{r}\right) & 0 & 0 & 0 \\ 0 & \left(1 - \frac{2M}{r}\right)^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2\theta \end{pmatrix}.$$

However, adding such a mass makes the system much more complicated. Luckily there are a few useful properties we can take advantage of. From Proposition 2.66 of Lee's *Introduction to Riemannian Manifolds*, we know that if (M, g) is a pseudo-Riemannian manifold, then for each p in M , there exists a smooth orthonormal frame on a neighborhood of p in M [5]. When we add mass to spacetime and deviate from the Minkowski metric, we can use a similar proposition. albeit not as strong. Instead we have that for any point p in spacetime, there is a coordinate frame such that $g_{\mu\nu}(p) = \eta_{\mu\nu}$ with $\eta_{\mu\nu}$ being the Minkowski metric [8]. We cannot necessarily do this for a neighborhood around p anymore, only exactly at the point p .

C. Christoffel Symbols

Additionally, Christoffel symbols are quite useful in general relativity, mainly to find the equations of motion

for an object in a particular geometry. To do so, we use the *geodesic equation* given by

$$\frac{d^2 x^\alpha}{d\tau^2} = -\Gamma_{\beta\gamma}^\alpha \frac{dx^\beta}{d\tau} \frac{dx^\gamma}{d\tau}$$

where τ is the proper time measured by a clock traveling along a geodesic [9]. This equation makes sense because for flat Minkowski space, all the Christoffel symbols will be zero. This means that an observer following geodesic in Minkowski space is stationary in space while moving forward in time. Additionally, to find the Christoffel symbols, we can use the equation

$$\Gamma_{\mu\nu}^\sigma = \frac{1}{2} g^{\sigma\rho} (\partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu})$$

with an implicit sum over ρ [8]. This equation allows us to find the Christoffel symbols for higher dimensions, not just for surfaces. Even though this equation and the ones used in class look quite different, we can see that this equation does give us the same results. For example, in spherical geometry with a first fundamental form of $R^2 d\theta^2 + R^2 \sin^2 \theta d\phi^2$ for constant R , we can find the Christoffel symbol $\Gamma_{\theta\phi}^\phi = \cot \theta$ using both formulas.

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